Job Loss and Coping Strategies in Sub-Saharan Africa: Evidence from Custom Labor Force Surveys in Senegal

Abdoulaye Cisse[a,1] **and Abdoulaye Ndiaye**[b,1,2]

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Drawing on rich panel labor force surveys and custom worker behavior surveys in Senegal, we uncover how employment formality and income shape coping mechanisms following job loss in a highly informal labor market. Wage earners, particularly formal workers, support large networks of unemployed and inactive individuals, yet they navigate a volatile labor market with high employment transitions and low job security. Informal self-employed workers experience the highest turnover, facing frequent job changes and, on average, a 40% drop in household consumption upon job loss. Despite these challenges, most wage earners, especially poorer informal workers, lack strong coping mechanisms, struggling to pay outstanding loans and bills or access reliable financial support. While formal workers experience a larger proportional consumption loss due to their financial responsibilities, informal workers have fewer safety nets and limited access to credit, leaving them highly vulnerable. Covering a long enough period to capture labor market shifts, including the COVID-19 shock, these surveys provide rare insight into employment volatility and worker resilience and reveal a precarious dynamic: while formal workers face significant financial pressure from their dependents, informal workers, especially those who are poor, have the least ability to absorb income shocks. Recognizing these differences is critical for designing broad-based policies that strengthen employment protections, expand social safety nets, and help reduce poverty risks in regions characterized by high informality, high employment volatility, high reliance on social networks, and limited social protections.

Sub-Saharan Africa | Senegal | job loss | unemployment | coping strategy | informality

Significance Statement

Workers in Sub-Saharan Africa face severe challenges when they lose their jobs, yet responses differ sharply by employment type and income. Our research from Senegal shows that formal workers experience a larger proportional drop in consumption upon job loss, suggesting that more people depend on their earnings. Still, they have stronger coping mechanisms, such as access to credit and safety nets. In contrast, poor informal workers report a smaller share of consumption loss, likely due to limited discretionary spending, but they also have far fewer coping strategies—even within informal networks, leaving them highly vulnerable. These findings highlight the importance of strengthening labor protections and expanding social safety nets to mitigate economic shocks and support poverty reduction efforts.

Poverty remains a persistent challenge in many regions of the world, particularly in Sub-Saharan Africa, where a large proportion of the population struggles to secure stable income sources and adequate living conditions. Despite notable efforts by governments, NGOs, and international development organizations, vulnerable households continue to grapple with economic uncertainties that threaten their livelihoods. One of the most pressing concerns in the region is the high prevalence of informal employment, which exists alongside a more structured formal labor sector. While the informal sector often offers flexibility and a degree of economic resilience in times of crisis, it also lacks the social protections and income stability typically associated with formal employment. These dual labor markets highlight the need to study how workers adapt to income shocks, what coping mechanisms they rely upon, and how policymakers can tailor interventions that address both immediate necessities and long-term development objectives.

This paper addresses these issues by analyzing data collected from Senegal, a country whose labor market dynamics reflect broader patterns observed across Sub-Saharan Africa. Through the combined use of the Employment Survey (ENES) and the Informal Sector Survey (ERISI), we gain valuable insights into the distribution of employment by region, the structural composition of the labor market, and how individuals and households respond to financial hardships. Focusing on a timeframe from 2015 to 2019, we capture an evolving economic landscape shaped by domestic policies, global market shifts, and local community support structures.

We build on (1), which presents stylized facts about the Senegalese labor market. There, we use nationally representative living standards and labour force surveys to document five key characteristics of the Senegalese labour market. Most workers are informal, with only a small fraction of workers and firms being formalized. Even within formal firms, there is a substantial presence of undeclared informal workers. There are pronounced income, consumption, and asset disparities across different employment statuses and, therefore, high potential for consumption smoothing through social insurance. Even within formal employment settings, a majority of

Author affiliations: [a]University of California, Berkeley, Department of Agricultural and Resource Economics, Berkeley, CA 94705 USA. https://orcid.org/0009-0006-4924-2475; [b]New York University, Stern School of Business, Henry Kaufman Management Center, Office 7-72, 44 West 4th, New York, NY 10012. https://orcid.org/0009-0000-7466-6444

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[1] AC contributed equally to this work with AN.

[2]To whom correspondence should be addressed. E-mail: abdou.ndiaye@nyu.edu.

workers lack significant work-related benefits, highlighting the need for a more robust social safety net. Informal networks are a significant form of insurance for workers.

Our analysis integrates descriptive statistics and econometric modeling to map out the determinants of employment stability in various regions and industries. By highlighting the role of informal employment in buffering individuals against economic downturns, we also illustrate the potential initiatives aimed at increasing informal sector incomes to act as stepping stones toward more secure livelihoods. At the same time, we highlight the inherent vulnerabilities that arise when social protections and legal frameworks are weak or absent—situations that can exacerbate the cycle of poverty for many households.

This research aims to inform policy discussions by offering a more granular view of how people navigate financial insecurity and employment transitions in impoverished contexts. In doing so, we hope to contribute practical recommendations for supporting micro-enterprises, strengthening community-based financial systems, and developing vocational training programs that align with the realities of the informal sector. The findings and policy implications herein serve as a call to action for national governments, international bodies, and civil society organizations committed to promoting inclusive growth and sustainable development in Sub-Saharan Africa.

Related Literature. This paper is related to the literature on labor market dynamics in low-income countries. Workers in low-income countries transition frequently to and from marginal employment without permanently moving to better-paying jobs, as documented in (2). In addition, nominal wage rigidity is particularly strong in village economies of low-income countries (3), which speaks to the slow rise in their living standards.

A broad literature examines how households cope with risk and the absence of formal insurance. In environments where formal unemployment or welfare benefits are lacking, families rely on informal arrangements such as mutual support networks, savings, and asset sales to smooth consumption. Early studies in South Asia demonstrated that rural families use strategies like marrying into different villages or temporary migration to diversify risk across locations (4). In sub-Saharan Africa, informal credit and reciprocal transfers serve a similar insurance role: for instance, credit transactions in northern Nigeria effectively pool risk between borrowers and lenders when incomes fluctuate (5). At the same time, these informal insurance mechanisms are far from complete. Studies of African households experiencing major shocks find that consumption often drops significantly, indicating only partial insurance. In rural Burkina Faso, households facing a drought were able to smooth consumption very little, despite using grain stocks and livestock sales—pointing to serious limitations of informal risk-sharing and self-insurance (6). The inability of communities to fully insure against aggregate shocks has important consequences for poverty and vulnerability.

Migratory labor is another important coping mechanism highlighted in the literature. (4) shows how sending a family member to work elsewhere (for example, an urban area) can allow rural households to diversify income sources and buffer agricultural shocks. Experimental evidence reinforces the value of migration in mitigating risk: (7) find that facilitating seasonal migration from famine-prone rural areas in Bangladesh led to higher earnings and improved food security for source households during the lean season.

Finally, the literature has evaluated the role of government interventions in relaxing the constraints faced by vulnerable workers. Studies of cash transfer programs (8, 9) illustrate the importance of informal sharing: (10) document that in Mexico's Progresa program, even households who were not direct beneficiaries experienced consumption gains because recipient households shared some of their transfer income with neighbors and relatives. Such spillover effects show that informal support networks remain active even when formal assistance is introduced, and they form a critical part of how communities respond to shocks and policy changes.

Results

Panel A and Panel B of Figure 1 track worker status for the same individuals in the EHCVM across 5 years. Panel A illustrates the five-year employment transitions across formal, informal, unemployed, and inactive categories, providing insight into labor market dynamics before and after the COVID-19 pandemic. The results highlight significant labor mobility across all employment statuses, particularly among informal workers who experience substantial shifts in employment status over time. Many individuals who were informally employed in 2018–2019 did not remain in the same category by 2021–2022, with some transitioning into unemployment, inactivity, or even formal employment. The high turnover in informal employment suggests the precarious nature of such jobs, where job stability is often low, and workers face greater exposure to economic shocks. Even among formal workers, there is notable movement, reflecting job losses and employment restructuring over the five-year period. Unemployed individuals and inactive populations also display substantial movement, with some re-entering the labor market, highlighting the fluidity of employment participation. The data captures both upward and downward mobility, indicating that employment transitions are not one-directional. It is particularly important to consider that this period includes the COVID-19 pandemic, which led to severe economic disruptions, disproportionately affecting informal workers and increasing job insecurity across all employment categories.

Panel B of Figure 1 extends this analysis by examining transitions across employment types (self-employed, employees, unemployed, inactive). The results suggest that self-employed individuals experienced particularly high volatility, with a large share shifting into unemployment or wage employment over the five-year period. This aligns with the challenges faced by self-employed individuals during economic downturns, where limited access to social protections and credit constraints can force them to exit self-employment. Conversely, employees—who are primarily wage earners—also exhibit notable transitions, with many moving into unemployment or informal self-employment, reflecting the fragility of job security, particularly in times of economic uncertainty. The large movement in employment status across all categories further reinforces the dynamic nature of the labor market, where even smaller groups of workers experience considerable change over time. The disruptive effects of COVID-19 are likely a major contributing factor to these employment shifts, with many workers being forced out of wage employment and into informal arrangements, unemployment, or inactivity.

Panel C of Figure 1 presents data on worker confidence in job security across formal and informal employment between 2017 and 2019, offering key insights into perceptions of labor market stability. A consistent trend emerges where formal workers report significantly higher confidence in keeping their jobs over the next twelve months compared to informal workers. In all years, a substantial majority of formal workers express confidence in job retention, with over 70% responding "Yes" to the question. This is in stark contrast to informal workers, where confidence is notably lower. In 2017, less than 45% of informal workers expressed confidence in job retention, while approximately 34% responded "No" and 21% responded "I don't know". Over time, job confidence among informal workers fluctuates but remains consistently lower than that of formal workers. By 2019, only 52% of informal workers express confidence in keeping their jobs, with a significant share (32%) still uncertain about their job security. Meanwhile, for formal workers, the proportion of those responding "Yes" remains relatively stable, around 70%, though there is a slight increase in uncertainty in 2019. The differences between formal and informal workers highlight the precarity of informal employment, where job stability is less certain, and workers face higher risks of displacement. The persistence of a sizable share of workers responding "I don't know" also suggests that job insecurity remains a concern even for those who are currently employed, particularly among informal workers. Over the three-year period, while formal worker confidence remains high and relatively stable, informal workers continue to experience higher uncertainty and lower perceived job stability, reinforcing the structural vulnerabilities of informal employment in the labor market.

Figure 2 provides a visualization of the primary coping mechanisms used by different age groups (Young, Middle-Aged, and Old) in both urban and rural settings. The results highlight significant reliance on family support across all demographics, though its extent varies by age and location.

Among young individuals, family is the overwhelmingly dominant coping mechanism, with over 80% of urban youth and nearly 79% of rural youth relying on family for financial support. Savings play a secondary role, with 16% of urban youth and 16% of rural youth reporting the ability to rely on personal savings. Other mechanisms, such as investment or pensions, are rarely used, with less than 4% of rural youth and just under 2% of urban youth indicating reliance on investments. This suggests that young people, particularly in urban areas, have few independent coping strategies and remain highly dependent on family networks.

Among middle-aged individuals, family remains the most relied-upon coping strategy but to a lesser extent than among younger individuals. Urban middle-aged individuals rely on family at a rate of 58%, while rural middle-aged individuals report a lower but still substantial rate of 54%. Unlike young individuals, savings play a larger role, accounting for 30% of urban middle-aged workers and 37% of their rural counterparts. Investments are used slightly more than in the younger group, particularly in rural areas, where 7% of middle-aged individuals rely on investments. Interestingly, middle-aged individuals are the only group where a small fraction report pensions as a coping strategy (0.2% in urban areas and 0.15% in rural areas), though at very low levels. This suggests that as individuals move into middle age, they diversify their financial safety nets beyond just family, but family remains their primary fallback.

Among older individuals, family reliance decreases further but remains the most frequently reported coping mechanism. Urban older individuals report a 49% reliance on family, while rural older individuals report 44%. Savings are particularly important for older individuals, with 42% of rural elderly and 30% of urban elderly depending on savings. Investment reliance increases with age, with 5% of urban elderly and 11% of rural elderly indicating investments as a coping strategy. Notably, pensions begin to play a small but significant role for older individuals, with 8% of urban elderly and 0.6% of rural elderly relying on pensions. This pattern suggests that while older individuals attempt to shift toward more stable financial resources such as savings and pensions, family remains the dominant safety net, particularly in rural settings where formal retirement mechanisms are less accessible.

Panel E of Figure 2 shows the share of workers who serve as a main financial support for their households, segmented by contract type (informal vs. formal employment) and income group (poor vs. rich). The findings highlight significant reliance on wage earners for household expenses, with particularly high levels of dependence among formal workers. The graph shows that formal workers, both poor and rich, are more likely to be the primary support of their household compared to informal workers. Among formal workers, the proportion of both poor and rich individuals who support their household exceeds 75%, with overlapping confidence intervals suggesting relatively similar reliance across income groups. This suggests that holding a formal job, regardless of income level, is associated with a higher likelihood that an individual is financially responsible for their family. This could be due to the relative stability and predictability of formal employment, making these workers more reliable sources of income for their households.

In contrast, among informal workers, there is a stark disparity between poor and rich individuals. Poor informal workers are significantly less likely to be the main financial support for their household, with their share closer to 40-50%, whereas rich informal workers report a much higher level of household dependence, approaching 70%. This difference suggests that informal workers with higher incomes may take on more financial responsibility within their families, despite working in a less stable employment arrangement. However, the lower level of reliance on poor informal workers could indicate that these individuals have less earning capacity and thus contribute less to household income, forcing families to rely on other sources of support, such as extended family or informal networks.

The high overall dependence on workers, particularly formal workers, shows the critical role of wage earners in household financial stability. This suggests that job loss or income shocks affecting these workers could have widespread consequences, not just for individuals but for entire families. The results also highlight the economic burden placed on formal workers, who are often expected to support multiple dependents, which could contribute to financial stress even among higher-income individuals.

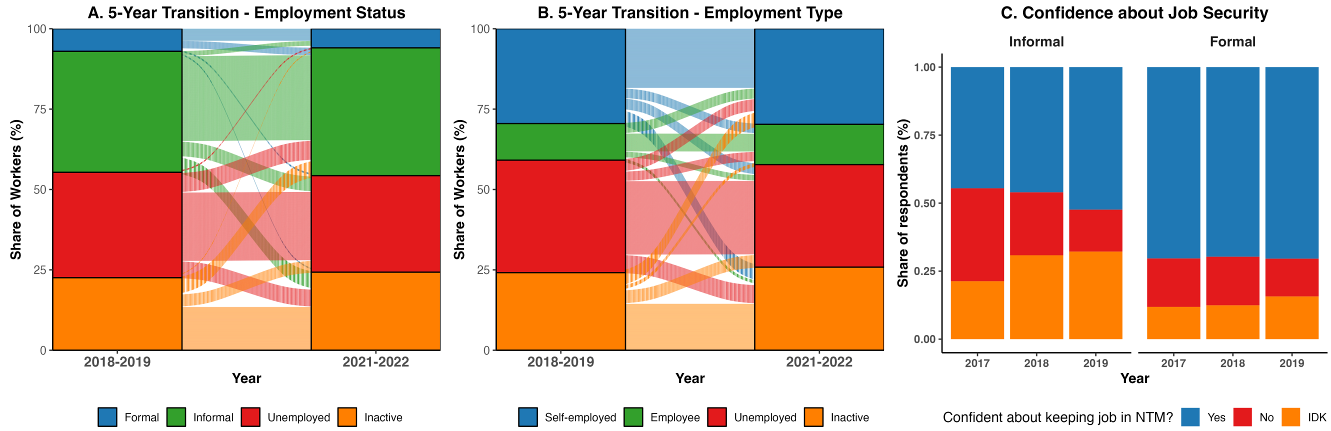

Fig. 1. High Transition Flows Across Employment Statuses and Low Worker Confidence about Job Security. Panel A and panel B use the EHCVM. Panel A tracks the employment status of respondents with different work status in the first round of the EHCVM (2018-2019) five years later in the second round of the EHCVM (2021-2022). Formal workers are wage workers with formal contracts. Informal workers are wage workers with no formal contracts. Unemployed individuals are people either looking for a job or not looking for a job due to involuntary reasons. Inactive people are people 15 years or older who are unemployed and not looking for a job for voluntary reasons. Panel B tracks the employment type of respondents in the first round of the EHCVM (2018-2019) five years later in the second round of the EHCVM (2021-2022). Employment type is as reported by the respondent. For Panel A and B, only respondents in the labor force (i.e. active people) are included in the analysis. Panel C uses the ENES data and displays the response to the question: "Are you confident about keeping your job in the net 12 months?", by formality type. "NTM" stands for next twelve months and "IDK" stands for I don't know. Both the EHCVM and the ENES data are described in the Data and Methods section.

Panel A of Figure 3 shows the drop in Consumption as a share of total household consumption upon job loss. According to Panel A, both rich–formal and rich–informal workers exhibit a larger percentage drop in total household consumption when faced with job loss compared to poorer households. This seemingly paradoxical finding can be explained by the fact that wealthier households, who generally consume more overall, can afford—and therefore expect—to cut a greater share of their spending if income suddenly disappears. In contrast, poorer households have less discretionary or "non-essential" consumption to reduce in the first place, so their expected drop as a fraction of total consumption appears smaller. Where standard errors do not overlap—for instance, when comparing rich–formal to poor–informal—the difference in these consumption-cutting strategies is statistically significant. Despite this larger percentage drop, having more resources overall (by virtue of both higher income and often more robust safety nets) still means that rich households are less likely to face dire hardships than the poor, even though the percentage cut in their spending is higher.

Panels B and C of Figure 3 relate to utility bills and loan repayment) and focus on more specific financial obligations—paying monthly utility bills and repaying outstanding loans. Overall, the ability to pay outstanding utility bills is low (around 40%) and the ability to pay back loans is even lower (less than 20%); and the difference across groups is not as stark as in panel A. The difference between rich–informal and poor-informal workers and between rich–informal and poor–formal is narrow, suggesting that formal contract status can partially offset a lower income level in terms of ability to pay back loans and outstanding bills. These findings reinforce that formal employment typically improves access to institutional protections (like severance or insurance), while higher income mitigates short-term cash flow disruptions.

Panels D and E of Figure 3 display access to formal and informal borrowing. They examine how likely workers are to borrow from formal institutions (banks, microfinance, etc.) versus informal networks (family, friends, or community groups) after job loss. The highest bar in Panel D belongs to poor–formal workers, whose standard errors do not overlap with those of poor–informal individuals, pointing to a significant difference in their access to official credit channels. Even poor–formal workers may rival or surpass rich–informal workers in Panel D, underscoring the distinct advantage conferred by a formal contract. In Panel E, although rich–formal respondents also lead in informal borrowing, the gap between them and other groups tends to be less stark, with some overlap in error bars. In other words, being formal consistently helps most with formal lending options, while informal borrowing might depend more on social or familial networks that are not strictly tied to income level or employment status.

Panel F of Figure 3 displays the monthly Savings as a share of salary and shows the percentage of salary saved each month. Here, rich–formal households stand out with notably higher saving rates than poor–informal households—once again, with non-overlapping error bars suggesting that the difference is statistically significant. Rich–informal also save more, on average, than either poor group, although the advantage over poor–formal can be modest if their error bars overlap. Taken together, these six panels demonstrate that despite experiencing a larger percentage drop in consumption (Panel A), wealthier individuals—especially those with formal contracts—are better able to cope with unemployment through higher savings, stronger capacity to pay bills and loans, and greater access to both formal and informal credit. Meanwhile, poorer households, especially in the informal sector, face relatively smaller consumption cuts in percentage terms simply because they lack discretionary spending but also contend with constrained resources to survive prolonged joblessness, highlighting a significant vulnerability gap in times of economic distress.

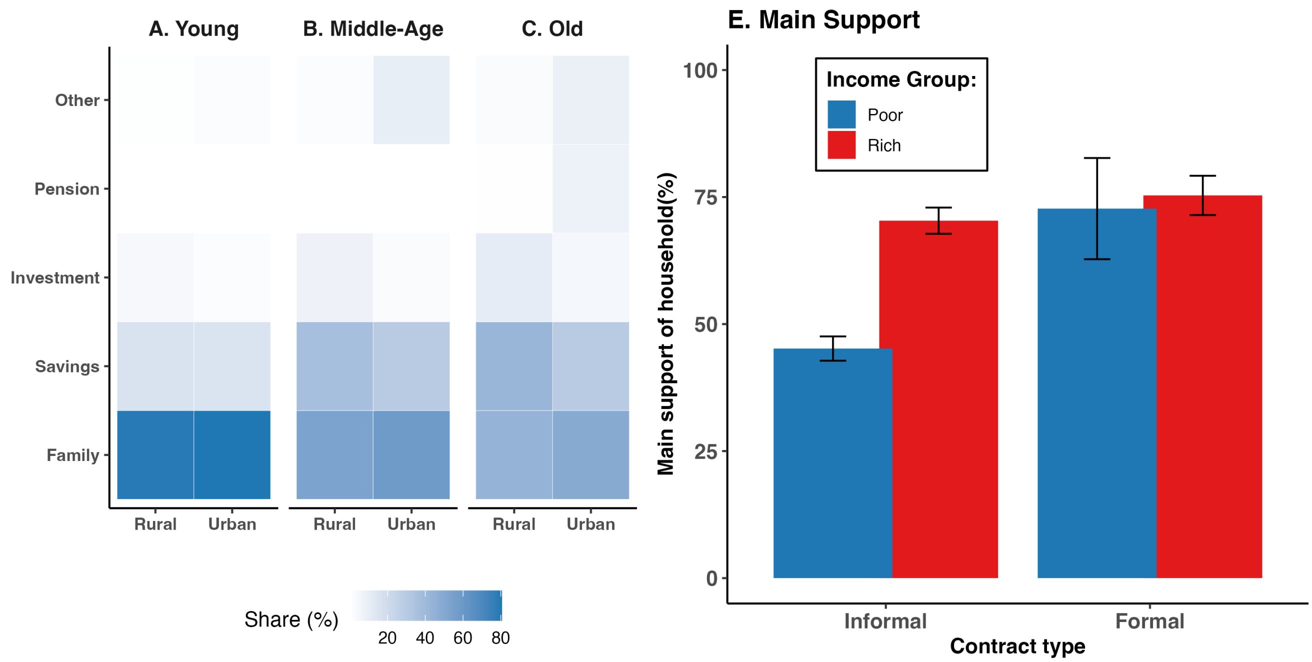

Fig. 2. High Dependence of Inactive and Unemployed Individuals on Family Wage Earners. The graph on the left-hand side of the figure uses all rounds of the ENES data described in the Data and Methods section (N = 132330). The figure restricts the analysis to individuals who are either inactive (out of the labor force) or are currently unemployed (i.e. looking for a job or not looking for a job due to involuntary reasons). Here, "Young" are individuals aged 25 or less, "Old" are those aged 55 or more and "Middle-Aged" are the rest. Each cell in the heat map gives the share of respondents who report relying on the coping mechanism listed on the same row of the cell on the y-axis as their main strategy to satisfy their basic needs at the time of the survey. Darker cells correspond to higher values while lighter cells correspond to lower values. The graph on the right-hand side of the figure is restricted to wage earners in our custom labor force survey described in the Data and Methods section. Individuals are categorized as poor vs. rich workers and formal vs. informal workers. Poor (rich) workers are workers whose salary is below (above) median salary. Formal workers are workers with a written contract or with employment benefits. The graph shows the share of wage workers who respond "Yes" to the question of whether they are the main provider of their family and the main support for household expenses (N=1378).

Discussion

Our findings contribute to and align with the above literature on employment vulnerability and coping strategies in low-income economies. The evidence from Senegal shows that a large fraction of workers lack stable formal employment and must navigate a precarious labor market. We observe that in the face of adverse shocks—such as an economic downturn or other disruption—affected workers frequently transition into informal jobs or self-employment as a fallback. This is consistent with the notion that the informal sector acts as a *buffer* during crises, absorbing workers who lose formal jobs. Indeed, the concept of "forced entrepreneurship" describes how individuals are pushed into self-employment not by choice but due to lack of wage jobs.

In line with previous studies, we find that households rely on and employ a variety of coping mechanisms to weather income shocks. However, while these strategies provide some short-term relief, they are not sufficient to fully prevent welfare losses. Our analysis of household consumption indicates that job loss often leads to a significant reduction in spending, highlighting the limits of informal coping. The lack of formal insurance or unemployment benefits (1) in the region leaves a gap that informal arrangements cannot entirely fill.

These insights have important policy implications. First, the prevalence of vulnerable employment suggests a need for policies that encourage the creation of more secure, formal jobs or improve the productivity and conditions of informal work. Reforms that reduce barriers to firm growth and formalization, or that support small enterprises, could gradually integrate more workers into the formal sector, as envisioned in dual-economy frameworks (11). Second, acknowledging wage rigidity and its impact, there may be benefits to interventions that promote labor market flexibility or provide alternatives to layoffs during downturns. For instance, work-sharing schemes or temporary wage subsidies in a crisis could preserve formal jobs and avoid pushing so many workers into informality. Finally, strengthening social protection is crucial. Given that households currently rely on informal networks for support, developing formal safety nets—such as unemployment insurance, cash transfer programs, or public works—could substantially reduce the welfare costs of job loss. Evidence from other settings demonstrates that even modest formal insurance can improve risk-taking and investment (12), which in turn may promote development. Additionally, formal transfers could complement, rather than fully displace, informal support systems (10), potentially enabling households to better manage shocks. Overall, our findings reinforce the argument that policies to address labor market frictions and enhance social protection would yield large benefits in Sub-Saharan Africa's context of pervasive job vulnerability.

Data and Methods

We use employment survey data spanning 2015–2019, focusing on: employment distribution by sector (formal/informal)

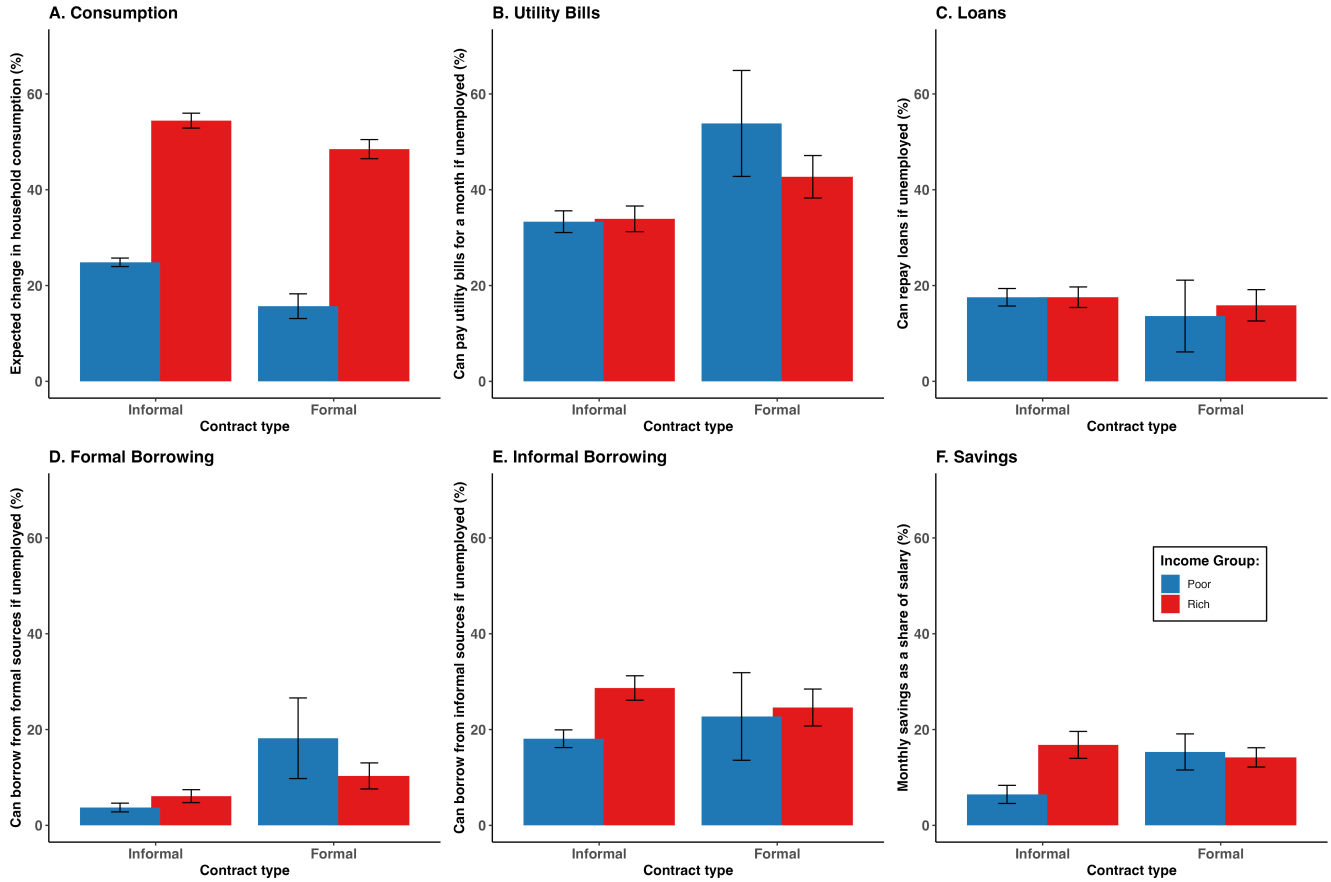

Fig. 3. Limited Coping Options upon Job Loss for Employed Individuals. Individuals are categorized as poor vs. rich workers and formal vs. informal workers. Poor (rich) workers are workers whose salary is below (above) median salary. Formal workers are workers with a written contract or with employment benefits. Panel A shows the change in household consumption as a share of current household consumption upon job loss. Panel B include both water and electricity bills. Panel C includes both formal and informal loans. Formal sources in Panel C refer to financial institutions officially registered. Panel D shows the share of workers accessing formal borrowing channels (e.g., banks, microfinance institutions) after job loss. Panel E illustrates the share of workers relying on informal borrowing sources (e.g., family, friends, community groups) following job loss. Panel F depicts the monthly savings (bank and mobile money savings, investments in real assets, and savings at home) as a percentage of salary for workers. All figures use the custom labor force survey described in the Data and Methods section. Only respondents employed at the time of the survey are included in the analysis. Vertical lines show standard errors.

and region, and household income shocks and response strategies, the role of informal employment in economic resilience. Data were analyzed using descriptive statistics and definitions of informality within the literature to understand the determinants of employment stability and adaptation to income shocks.

Data and Sample. First, we conducted a custom labor force survey with a representative sample of the urban population in Senegal. This approach aligns with common practices for labor force surveys conducted in low-income countries, which primarily focus on urban areas. Thus, we abstract from the spillover effects of labor market policies on rural migration emphasized in the literature (13, 14). Furthermore, rural areas, where agricultural workers are typically found in Senegal, have dedicated government programs, such as agricultural input subsidies, which could mitigate any potential effect of UI on rural–urban migration.

The design of the custom survey follows a stratified random sampling approach. First, we define the population of the study as all active workers, which are individuals aged 15 or above, in Dakar, the capital of Senegal. Second, we use enumeration areas (EAs) as our primary sampling units (PSUs), as defined by the national statistical agency during the 2013 population census of Senegal. These EAs are distributed across the five districts in the region of Dakar: Dakar district, Guediawaye, Keur Massar, Pikine, and Rufisque. We randomly select 23 EAs from the set of 129 in Dakar. Third, within each selected EA, we randomly sample 15 households. The survey thus covers all individuals aged 15 and over within these selected households. In total, we survey 1,378 individuals across 345 households.

Second, we use longitudinal household expenditure surveys conducted in Senegal in 2018/2019 and in 2021/2022, the Enquête Harmonisée sur les Conditions de Vie des Ménages 2018–2019 (EHCVM) and the Enquête Harmonisée sur les Conditions de Vie des Ménages 2021–2022, to incorporate in the analysis socioeconomic variables beyond those related to electricity that the datasets above provide. These surveys are very similar in spirit to the Living Standard Measurement Survey (LSMS) commonly conducted in several low-income countries. The data were collected by Agence Nationale de la Statistique et de la Démographie (ANSD) through a 2-stage sampling methodology. A total of 598 enumeration areas

(EAs) were selected in the first stage, and 12 households were randomly selected in each EA in the second stage. The total survey sample size is 7,156 households, with 3,941 from urban areas and 3,215 from rural areas. The EHVCM is a rich dataset covering education, health, employment, nonemployment income, saving and credit, food consumption, food security, nonfood consumption, nonagricultural enterprises, housing, household assets, transfers, shocks and survival strategies, safety nets, agriculture, livestock, fishing, agricultural equipment, and relative poverty. Importantly, I have the exact centroids of the EAs where respondents live. I use the centroids to match the EHCVM with the rest of the datasets.

Third, we used the *Enquête Nationale sur l'Emploi au Senegal 2015-2019* (ENES). The ENES consists of 12 waves of quarterly, nationally representative labor force surveys from between 2015 and 2019. The survey uses a rotating panel of households. The ENES covers (i) demographic information on education, gender, age, and family structure; (ii) information on employment status, contract structure, industry, occupation, earnings, working hours, formality type, tenure in the current job, and any changes in employment over the past three months; (iii) job search behavior with information on whether respondents engage in job search activities, the methods they employ in their job search, reasons for not actively seeking a job, and success in finding employment; (iv) consumption expenditures with information on the amount of money spent on food and beverages, utilities, and housing and any changes in these expenditures over the past few months; and (v) savings and borrowing with information on the methods used for saving and borrowing, the amount saved or borrowed, and whether the borrowing channels are formal or informal.

Fourth, we used the two waves of the Enquête régionale intégrée sur l'emploi et le secteur informel, which is representative survey of the informal sector.

Methods. We analyze our longitudinal household survey data segment workers across age (young, middle-age, old), geography (urban, rural), social status (poor, rich), and work status (formal, informal, unemployed). Poor (rich) workers are workers whose salary is below (above) median salary. Informal employment is defined as work lacking a formal contract or social security coverage, consistent with standard definitions used in the literature (11). For example, self-employed persons in small household enterprises and workers paid in cash without registration are classified as informal. By contrast, formal employment refers to wage jobs with a contract, including public sector and corporate jobs that typically provide benefits. We focus on the non-agricultural workforce in our custom labor force survey, but include agricultural labor in the total ENES and ERISI.

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